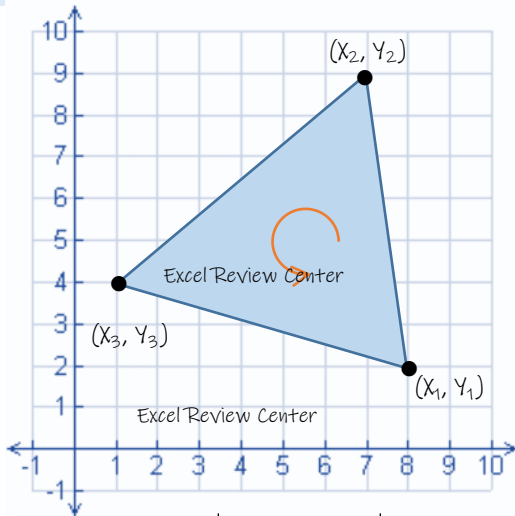




Area by Coordinates

Consider a polygon whose vertices have coordinates of (x_1, y_1) , (x_2, y_2) and (x_3, y_3) .



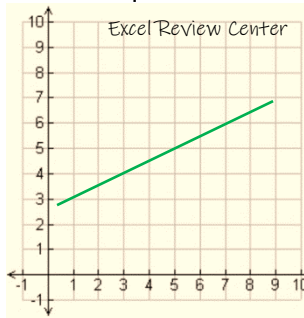
$$A = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & \dots & x_1 \\ y_1 & y_2 & y_3 & \dots & y_1 \end{vmatrix}$$

$$A = \frac{1}{2} [(x_1 y_2 + x_2 y_3 + x_3 y_1) - (y_1 x_2 + y_2 x_3 + y_3 x_1)]$$

Equations of Lines

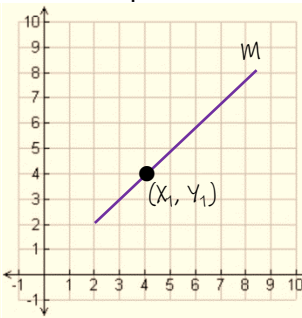
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General Equation



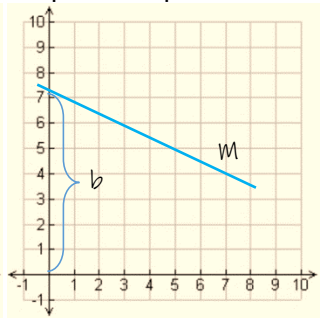
$$Ax + By + C = 0$$

Point-Slope Form



$$y - y_1 = m(x - x_1)$$

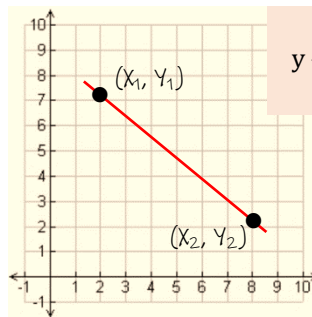
Slope-Intercept Form



$$y = mx + b$$

Two-Point Form

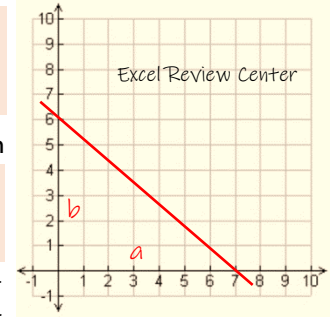
$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$



Intercept Form

$$\frac{x}{a} + \frac{y}{b} = 1$$

a = x-intercept
 b = y-intercept



Conic Sections

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Parabola

Circle

Ellipse

Hyperbola

General Equation of Conic Sections

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

When B is not equal to zero, then the principal axes of the conic are inclined (not parallel to the coordinates axes). The curve can be identified from the equation given by determining the value of the determinant, $B^2 - 4AC$ to intersect a cone.

$B^2 - 4AC$	Conic Section	Eccentricity
< 0	Ellipse	< 1.0
$= 0$	Parabola	$= 1.0$
> 0	Hyperbola	> 1.0

When B is equal to zero, then the principal axes of the conic are parallel to the coordinates axes (x and y axes). To identify the curve, compare the coefficients of A and C .

Conic section (or simply Conic) is the locus of a point which moves so that its distance from a fixed point (focus) is in constant ratio, e (eccentricity) to its distance from a fixed straight line (directrix). The term "conic" was first introduced by a renowned mathematician and astronomer of antiquity, Apollonius (c.255 – 170 B.C.). Also, the term "conic section" was due to the fact that the section is formed by a plane made to intersect a cone.

If $A = C$, the conic is a circle.

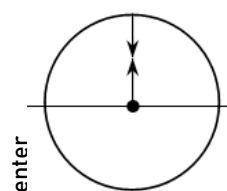
If $A \neq C$ but the same signs, the conic is an ellipse.

If A and C have different signs, the conic is a hyperbola.

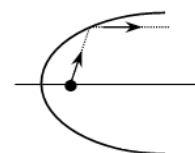
If either A or C is zero, the conic is a parabola.

Geometric Properties of Conic Sections

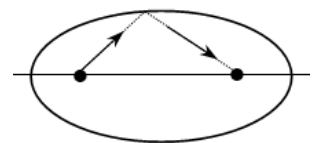
The conic sections have geometric properties that can be used for some engineering application such as beams of sound and reflection of rays of light.



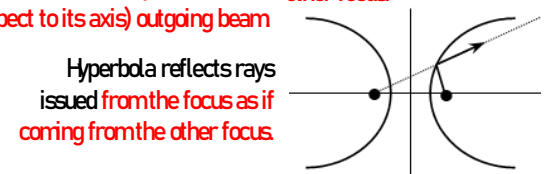
Circle reflects rays issued from the focus back to the center of the circle.



Parabola reflects rays issued from the focus as a parallel (with respect to its axis) outgoing beam



Ellipse reflects rays issued from the focus into the other focus



Hyperbola reflects rays issued from the focus as if coming from the other focus